# Edexcel A Level Physics

Your notes

## Radioactivity Decay

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- Background Radiation
- Alpha, Beta & Gamma Radiation
- Nuclear Decay Equations
- Core Practical 15: Investigating Gamma Radiation Absorption
- The Random Nature of Nuclear Decay
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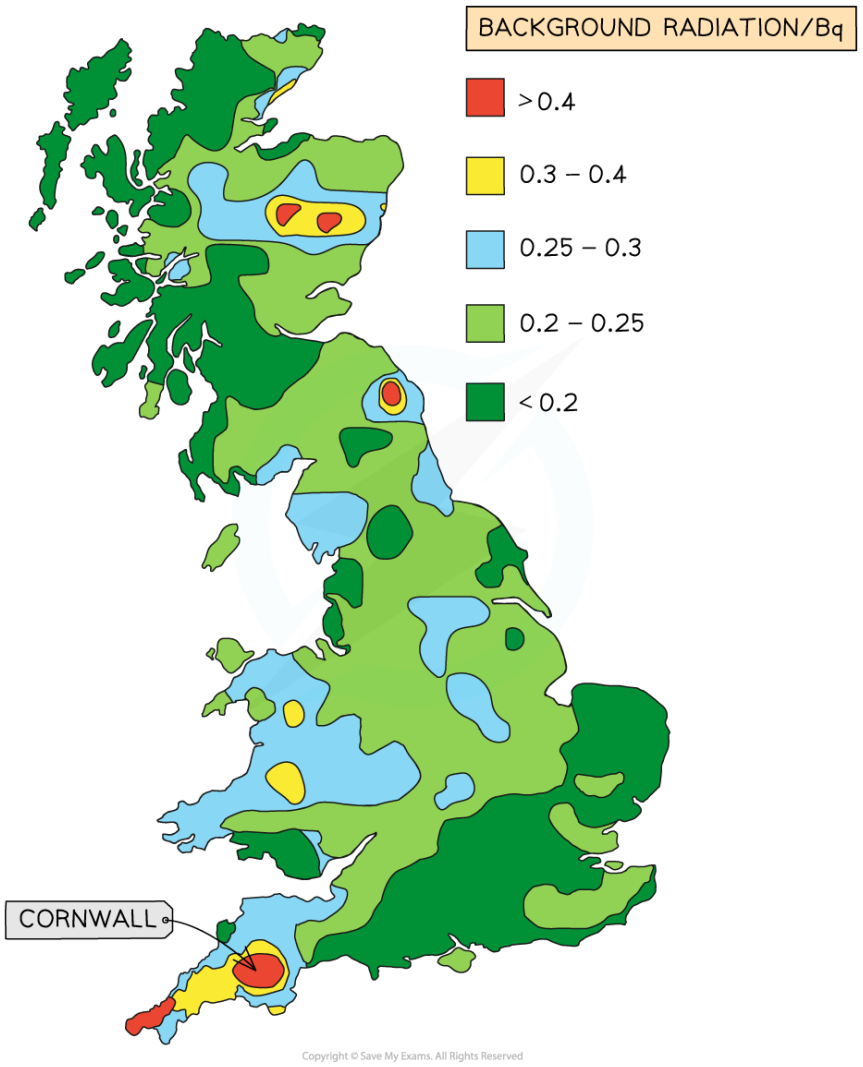
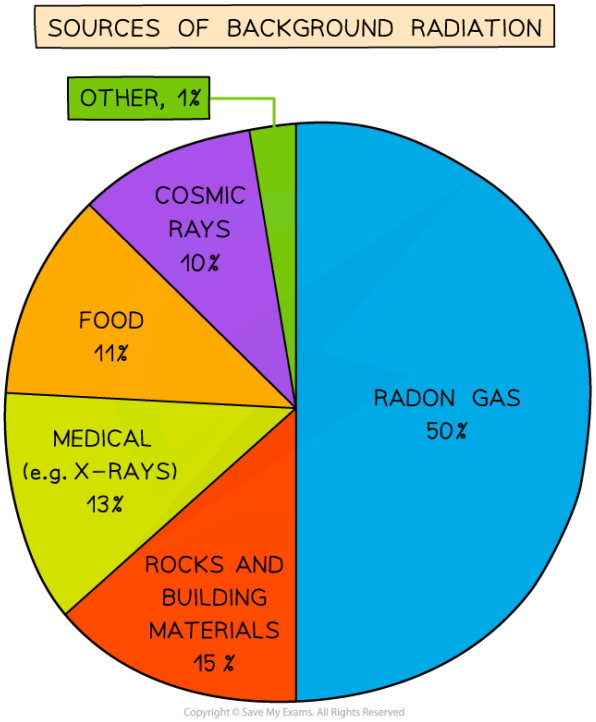
# Background Radiation

- Radiation is a natural phenomenon, with radioactive elements having always existed on Earth and in outer space
  - However, human activity has added to the amount of radiation that humans are exposed to in various ways
- Background radiation is defined as:

  **Low levels of radiation from environmental sources, which are always present around us**
- Radiation is measured in **counts per second** in a unit called **Becquerel (Bq)**
- Different amounts of radiation are present in different places around the world, including in the UK.



- There are two types of background radiation:
  - Natural sources
  - Man-made sources




***Background radiation is the radiation that is present all around the environment. Radon gas is given off from some types of rock***

- Every second of the day there is some radiation emanating from **natural sources** such as:
  - Rocks
  - Cosmic rays from space
  - Foods

## Natural Sources

- **Radon gas from rocks and soil**
  - Heavy radioactive elements, such as uranium and thorium, occur naturally in rocks in the ground
  - Uranium decays into radon gas, which is an alpha emitter
  - This is particularly dangerous if inhaled into the lungs in large quantities
- **Cosmic rays from space**
  - The sun emits an enormous number of protons every second

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- Some of these enter the Earth's atmosphere at high speeds
- When they collide with molecules in the air, this leads to the production of gamma radiation
- Other sources of cosmic rays are supernovae and other high energy cosmic events

- **Carbon-14 in biological material**
  - All organic matter contains a tiny amount of carbon-14
  - Living plants and animals constantly replace the supply of carbon in their systems hence the amount of carbon-14 in the system stays almost constant
- **Radioactive material in food and drink**
  - Naturally occurring radioactive elements can get into food and water since they are in contact with rocks and soil containing these elements
  - Some foods contain higher amounts such as potassium-40 in bananas
  - However, the amount of radioactive material is minuscule and is not a cause for concern

## Man-Made Sources

- **Medical sources**
  - In medicine, radiation is commonly used in X-rays, CT scans, radioactive tracers, and radiation therapy
- **Nuclear waste**
  - While nuclear waste itself does not contribute much to background radiation, it can be dangerous for the people handling it
- **Nuclear fallout from nuclear weapons**
  - Fallout is the residue radioactive material that is thrown into the air after a nuclear explosion, such as the bomb that exploded at Hiroshima
  - While the amount of fallout in the environment is presently very low, it increases significantly in areas where nuclear weapons are tested
- **Nuclear accidents**
  - Accidents such as that in Chernobyl contributed a large dose of radiation into the environment
  - While these accidents are now extremely rare, they can be catastrophic

## Corrected Count Rate

- Background radiation must be accounted for when taking readings in a laboratory
- This can be done by taking readings with no radioactive source present and then subtracting this from readings with the source present
  - This is known as the **corrected count rate**

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## Detecting Radiation

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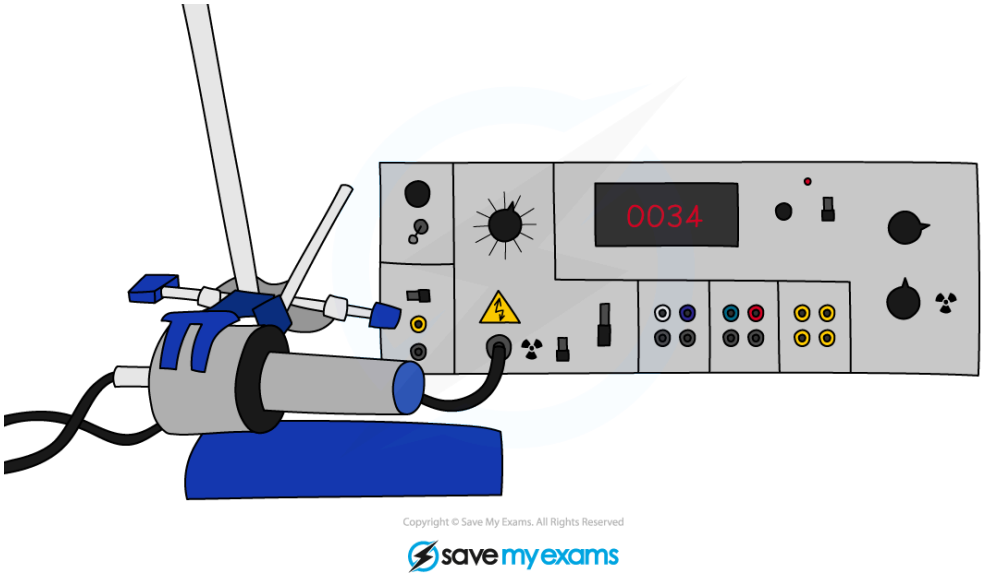
- When alpha or beta radiation pass close to an atom, they can deliver enough energy to remove electrons, ionising the atom
- Radiation detectors work by detecting the presence of either these ions, or the chemical changes that they produce
- Examples of radiation detectors include:
  - **Photographic film** (often used in badges)
  - **Geiger-Muller** (GM) tubes
  - **Ionisation chambers**
  - **Scintillation counters**
  - **Spark counters**




***A Geiger-Muller tube (or Geiger counter) is a common type of radiation detector***

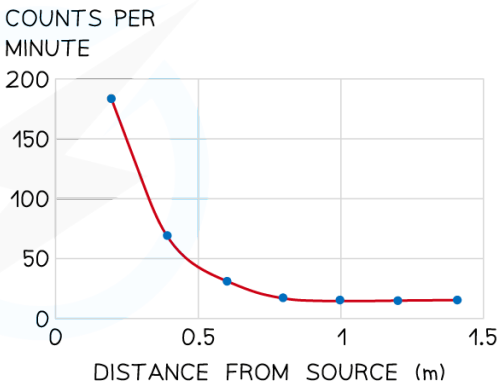
### Worked Example

A student is using a Geiger-counter to measure the counts per minute at different distances from a source of radiation. Their results and a graph of the results are shown here.

RESULTS TABLE

| Distance from source (m) | Counts per minute |
|---|---|
| 0.2 | 180 |
| 0.4 | 67 |
| 0.6 | 29 |
| 0.8 | 17 |
| 1.0 | 15 |
| 1.2 | 15 |
| 1.4 | 15 |

GRAPH

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Determine the background radiation count.

**Answer:**

**Step 1: Determine the point at which the source radiation stops being detected**

- The background radiation is the amount of radiation received all the time
- When the source and detector are far enough apart, the radiation is absorbed by the air before reaching the Geiger-counter
- Results after 1 metre should not change
- Therefore, the amount after 1 metre is only due to background radiation

**Step 2: State the background radiation count**

- The background radiation count is **15 counts per minute**

## Examiner Tips and Tricks

Exam questions may expect you to remember about the existence of background radiation without mentioning it. Look out for count rates that do not drop to zero, or half-life graphs with a line that tends towards a value higher than zero.

When memorising lists of the causes of background radiation, make sure to choose at least one natural and one man-made cause as these are thought of quite separately.

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# Alpha, Beta & Gamma Particles

- Some elements have nuclei that are unstable
  - This tends to be when the number of nucleons does not balance
- In order to become more stable, they emit particles and/or electromagnetic radiation
  - These nuclei are said to be **radioactive**
- There are three different types of radioactive emission: Alpha, Beta and Gamma

## Alpha Particles

- **Alpha (α) particles** are high energy particles made up of **2 protons and 2 neutrons** (the same as a helium nucleus)
- They are usually emitted from nuclei that are too large

$$^{4}_{2}\alpha \quad \text{OR} \quad ^{4}_{2}He$$

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- Alpha particles are the **most** ionising type of radiation
  - This is due to having the highest charge of +2e
  - They produce the highest number of ion pairs per cm in air (~10 000 ion pairs per cm)
  - This means they can do more damage to cells than the other types of radiation
- Alpha particles are the **least** penetrating type of radiation
  - This means they travel the shortest distance in air before being absorbed
  - They have a range of around 3 to 7 cm in air

## Beta Particles

- **Beta ($\beta^-$) particles are high energy electrons** emitted from the nucleus
  - $\beta^-$ particles are emitted by nuclei that have too many **neutrons**

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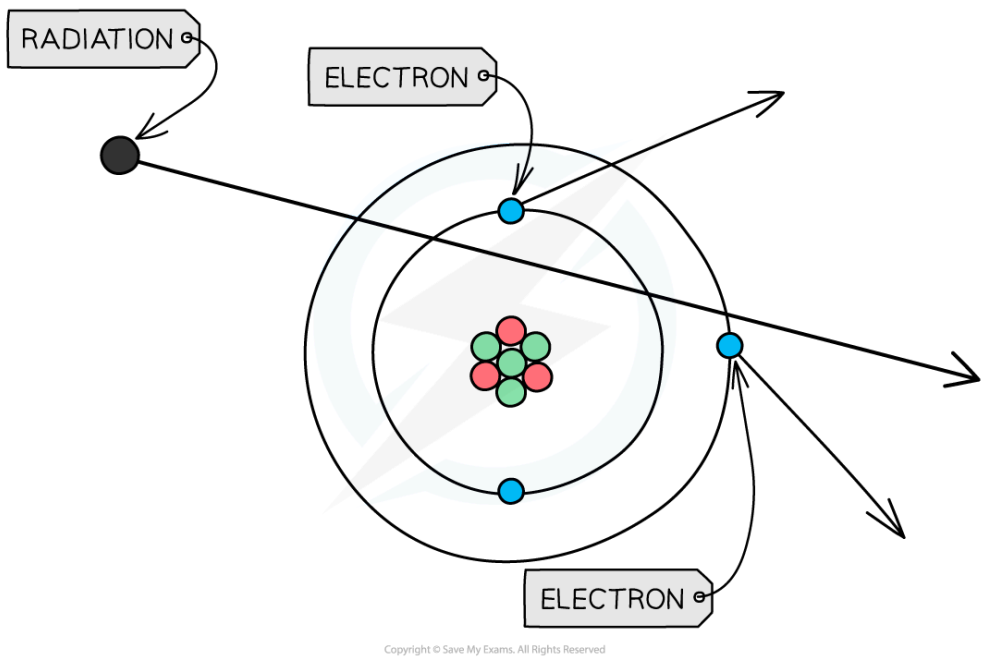
- Beta is a **moderately** ionising type of radiation
  - This is due to it having a charge of +1e
  - This means it is able to do some slight damage to cells (less than alpha but more than gamma)
- Beta is a **moderately** penetrating type of radiation
  - Beta particles have a range of around 20 cm - 3 m in air, depending on their energy
- Beta can be stopped by a few millimetres of **aluminium** foil

## Gamma Rays

- **Gamma (γ) rays are high energy electromagnetic waves**
- They are emitted by nuclei that need to lose some energy

- If these particles hit other atoms, they can knock out electrons, **ionising the atom**
- This can cause chemical changes in materials and can damage or kill living cells

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***When radiation passes close to atoms, it can knock out electrons, ionising the atom***

- The properties of the different types of radiation are summarised in the table below

| Particle | Composition | Mass / u | Charge / e | Speed / c |
|---|---|---|---|---|
| Alpha ($\alpha$) | 2 protons + 2 neutrons | 4 | +2 | 0.05 |
| Beta minus ($\beta^-$) | Electron ($e^-$) | 0.0005 | −1 | > 0.99 |
| Beta plus ($\beta^+$) | Positron ($e^+$) | 0.0005 | +1 | > 0.99 |
| Gamma ($\gamma$) | Electromagnetic wave | 0 | 0 | 1 |

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- *u* is the atomic mass unit (see "Atomic Mass Unit (u)")
- e is the charge of the electron: $1.60 \times 10^{-19}$ C
- c is the speed of light: $3 \times 10^{8}$ m s$^{-1}$

**Worked Example**

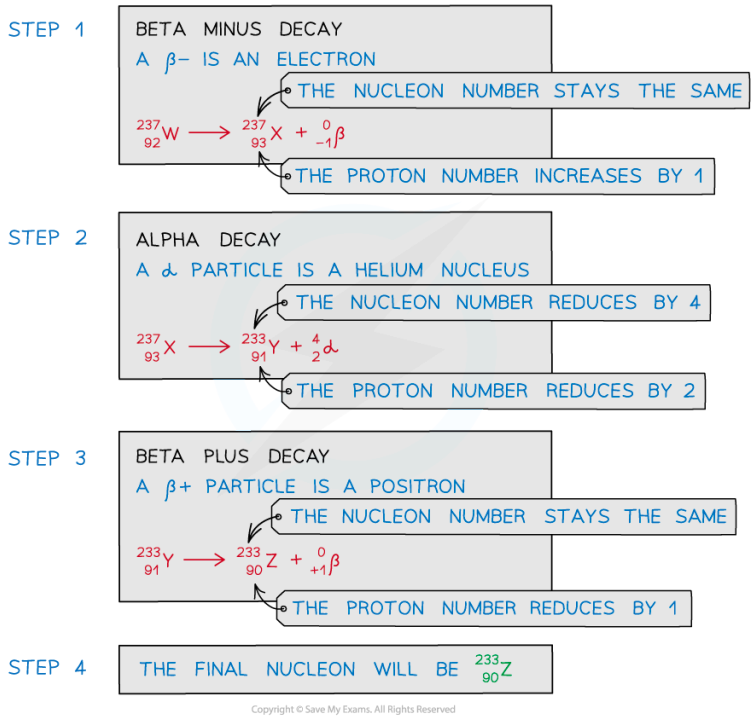
Three successive radioactive decays are shown in the diagram below; each one results in a particle being emitted.
The first decay results in the emission of β-particle. The second decay results in the emission of an α-particle. The third decay results in the emission of another β-particle.

$$^{237}_{92}W \xrightarrow{\beta^-} X \xrightarrow{\alpha} Y \xrightarrow{\beta^+} Z$$

Nuclides W and Z are compared.
Which nuclide of Z is formed at the end of this decay?

**A.** $^{237}_{90}Z$ **B.** $^{233}_{92}Z$ **C.** $^{237}_{89}Z$ **D.** $^{233}_{90}Z$

**Answer: D**




## Examiner Tips and Tricks

It is important to be familiar the properties of each type of radiation and their symbols.

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# Nuclear Decay Equations

## Alpha Decay

- An alpha particle consists of **2 protons and 2 neutrons**

  (It is emitted from large unstable nuclei)

- When an alpha particle is emitted from a nucleus:
  - The nucleus loses 2 protons:

    **The proton (atomic) number decreases by 2**

  - The nucleus loses 4 particles (nucleons) in total:

    **The nucleon (mass) number decreases by 4**

- Equation for alpha emission:

$$^{A}_{Z}X \longrightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}\alpha$$

$$\text{e.g.} \quad {}^{212}_{84}Po \longrightarrow {}^{208}_{82}Y + {}^{4}_{2}\alpha$$

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- **Nuclear equations, just like chemical equations, balance**:
  - The sum of the upper (mass) numbers on the left of each equation should equal the sum on the right
  - The sum of the lower (atomic) numbers should also balance

## Beta Emission

- Equation for beta emission:

$$^{A}_{Z}X \longrightarrow {}^{A}_{Z+1}Y + {}^{0}_{-1}\beta$$

$$\text{e.g.} \quad {}^{14}_{6}C \longrightarrow {}^{14}_{7}N + {}^{0}_{-1}\beta$$

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- Note that the beta particle is given an atomic number of ***-1*** in the above examples

  This is because the atomic number is being used to measure charge in this case:

  - Protons, being positive particles, have positive atomic numbers
  - Electrons, being negative, have a negative number

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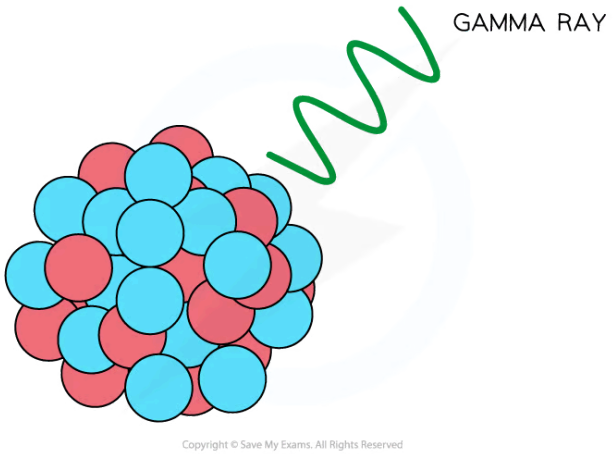
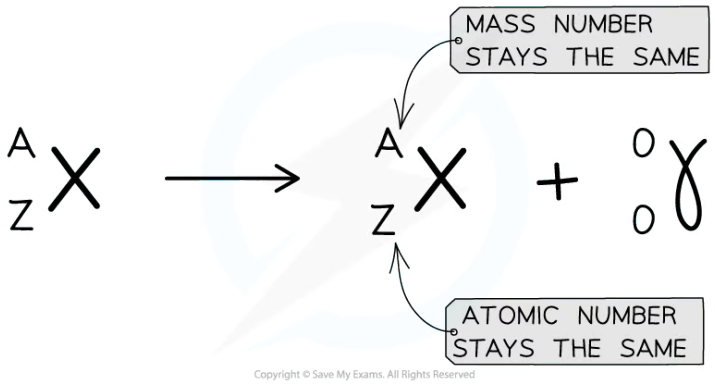
## Gamma Decay

- During gamma decay, a gamma ray is emitted from an unstable nucleus
- The process that makes the nucleus less energetic but does not change its structure




***Gamma decay does not affect the mass number or the atomic number of the radioactive nucleus, but it does reduce the energy of the nucleus***

- The gamma ray that is emitted has a lot of energy, but no mass or charge
- Here is an example of Uranium-238 undergoing gamma decay
  - Notice that the mass number and atomic number of the unstable nuclei remains the same during the decay




***Gamma decay equation***

$$^{A}_{Z}X \longrightarrow {}^{A}_{Z}X + \gamma$$

$$\text{e.g. } {}^{137}_{56}Ba \longrightarrow {}^{137}_{56}Ba + \gamma$$

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*An example of Barium decay through the release of a gamma ray*

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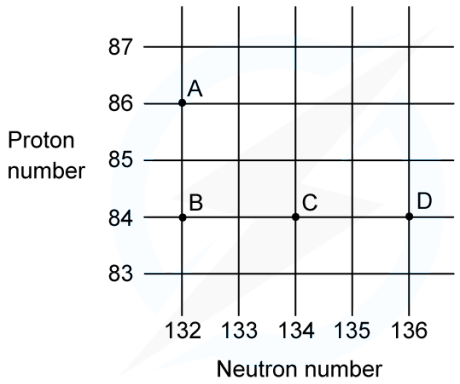
## Worked Example

The radioactive nucleus $^{222}_{86}Rn$ undergoes alpha decay into a daughter nucleus Po.




a.) Which letter in the diagram represents the daughter product?
b.) Which is the nucleon and proton number of the final nucleus Po?

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**Part (a)**

**Answer: C**

a.)

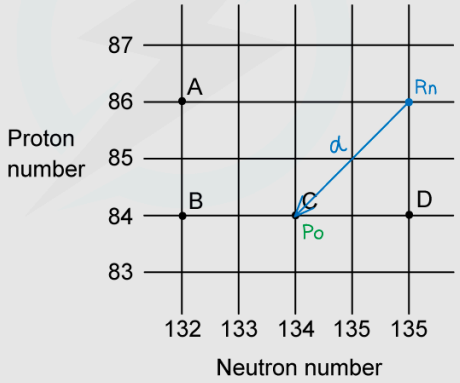
STEP 1 IN ALPHA DECAY, THE NUCLEON NUMBER OF THE PARENT NUCLEUS DECREASES BY 4 AND THE PROTON NUMBER DECREASES BY 2

STEP 2 THE NUMBER OF NEUTRONS IS:

$222 - 86 = 136$

STEP 3 A HELIUM NUCLEUS HAS 2 PROTONS AND 2 NEUTRONS

STEP 4 THE PROTON AND NEUTRON NUMBER BOTH DECREASES BY 2 REACHING POINT C

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Part (b)

b.)

STEP 1 ALPHA DECAY EQUATION:

$^{A}_{Z}X \longrightarrow {}^{4}_{2}\alpha + {}^{A-4}_{Z-2}Y$ $\quad$ $^{222}_{86}Rn \longrightarrow {}^{4}_{2}\alpha + {}^{218}_{84}Po$

STEP 2 NUCLEON NUMBER OF Po = 218

PROTON NUMBER Po = 84

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# Core Practical 15: Investigating Gamma Radiation Absorption

## Aim of the Experiment

- To investigate the absorption of gamma rays by different thicknesses of lead

## Variables:

- Independent variable = Thickness of lead
- Dependent variable = Count rate
- Control variables:
  - Radioactive source
  - Distance of GM tube to source
  - Location / background radiation

## Equipment List

| Equipment | Purpose |
|---|---|
| Radioactive gamma source | Used for testing |
| Ruler | To control the distance between the sources and the GM tu be |
| Mount for radioactive sources | To hold the sources securely |
| Geiger–Muller tube and counter | For measuring count rate |
| Tongs | For safe handling of the sources |
| Different thicknesses of lead | To be changed as the independent variable. Increasing intervals of 0.5 mm |
| Lead lined container for storing sources when not in use | For safety and to ensure results are not affected by sources not being tested |

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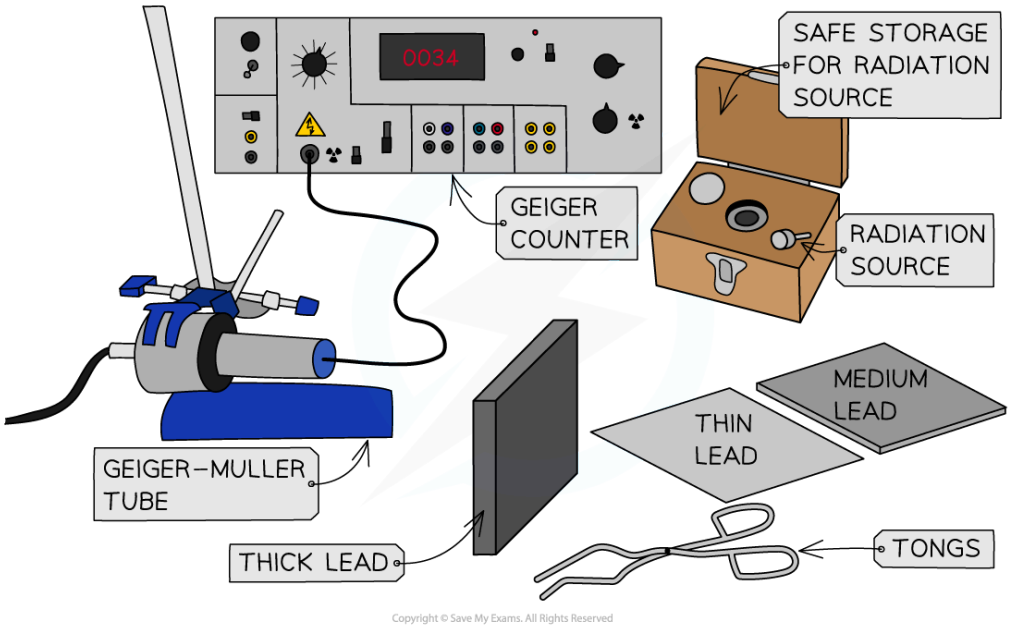
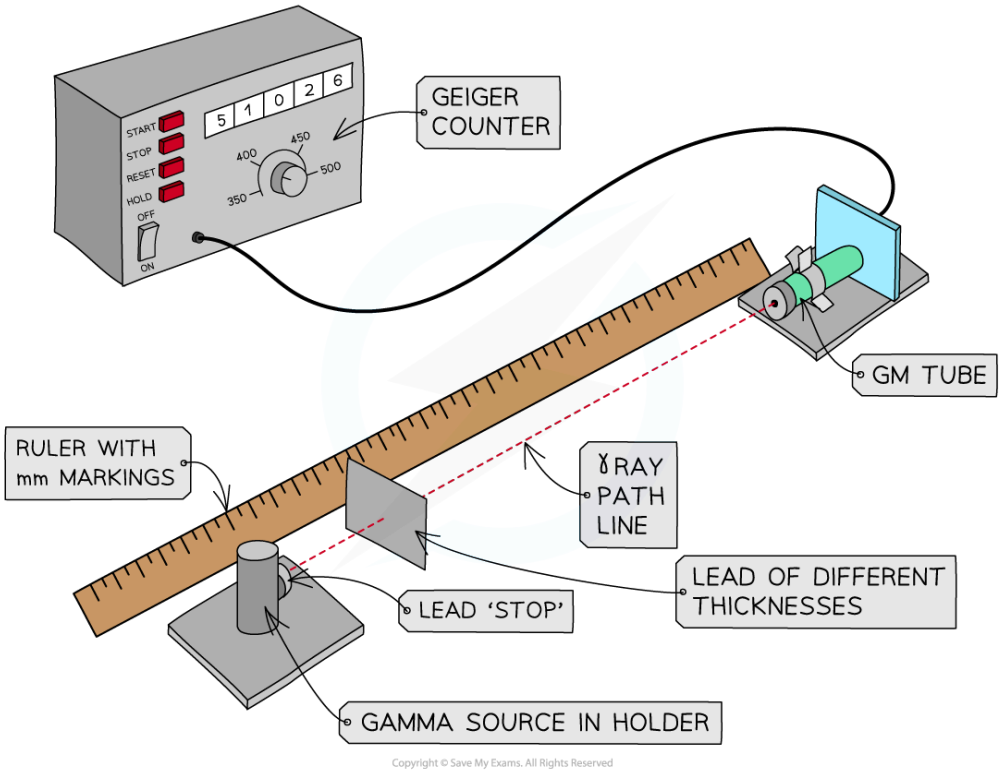



- Resolution of measuring equipment:
  - Ruler = 1 mm
  - Geiger-Müller tube = 0.01 μS/hr

## Method

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1. Connect the **Geiger-Müller tube** to the **counter** and, without any sources present, **measure background radiation** over a five-minute period
    - Record this value
    - Calculate the average background rate per minute
2. Measure the thickness of the lead absorbers using Vernier calipers at three points on each sheet.
    - For each sheet record the average thickness
3. Place the radioactive source **a fixed distance** of 10 cm away from the tube
4. Record the count rate over **one minute**
5. Repeat steps 3 and 4 a further two times, recording the count rate each time
6. Place the thinnest absorber directly in front of the gamma ray source
7. Repeat steps 3–5
8. Replace the sheet with another thickness and continue taking three readings per thickness

## Analysis of Results

- **If the count over that interval falls to background levels** (allow for a little random variation), then the radiation has all been absorbed
- You will be able to determine the thickness of the lead required to absorb gamma radiation

## Evaluating the Experiment

Systematic Errors:

- Make sure that the source is stored well away from the counter during the experiment
- Conduct all runs of the experiment in the same location to avoid changes in background radiation levels

Random Errors:

- The accuracy of such an experiment is improved with using a reliable source of radiation with a long half-life and an activity well above the natural background level

## Safety Considerations

- When not using a source, keep it in a lead-lined container
- When in use, try and keep a good distance (a metre or so) between yourself and the source
- When handling the source, do so using tweezers (or tongs) and point the source away from you
- Wash your hands and remove your outer layer of clothing after handling a radioactive source

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## Examiner Tips and Tricks

When answering questions about the core practicals you could try to remember the acronym SCREAMS:

- S: Which variable will you keep the **same**
- C: which variable should you **change**
- R: what will you do to make your experiment **reliable**
- E: what special **equipment and equations** are required
- A: how will you **analyse** your results
- M: which variable will you **measure**
- S: what **safety** precautions will you take?


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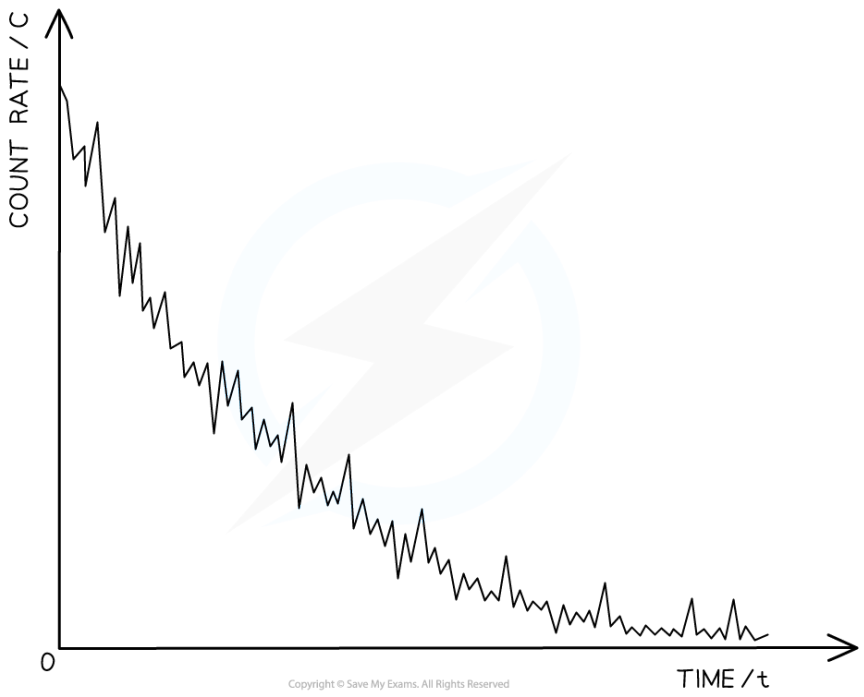
# The Random Nature of Nuclear Decay

- Radioactive decay is defined as:

  **The spontaneous disintegration of a nucleus to form a more stable nucleus, resulting in the emission of an alpha, beta or gamma particle**

- The random nature of radioactive decay can be demonstrated by observing the count rate of a Geiger-Muller (GM) tube
  - When a GM tube is placed near a radioactive source, the counts are found to be irregular and cannot be predicted
  - Each count represents a decay of an unstable nucleus
  - These fluctuations in count rate on the GM tube **provide evidence for the randomness of radioactive decay**




***The variation of count rate over time of a sample radioactive gas. The fluctuations show the randomness of radioactive decay***

- Radioactive decay is both **spontaneous** and **random**
- A spontaneous process is defined as:

  **A process which cannot be influenced by environmental factors**

- This means radioactive decay is not affected by environmental factors such as:

- Temperature
- Pressure
- Chemical conditions

- A random process is defined as:

  **A process in which the exact time of decay of a nucleus cannot be predicted**

- Instead, the nucleus has a **constant probability,** i.e.. the same chance, of decaying in a given time
  - Therefore, with large numbers of nuclei, it **is** possible to statistically predict the behaviour of the **entire group**

### Examiner Tips and Tricks

Make sure you can define what constitutes a radioactive decay, a random process and a spontaneous decay - these are all very common exam questions!

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# Activity & The Decay Constant

- Since radioactive decay is spontaneous and random, it is useful to consider the average number of nuclei which are expected to decay per unit time
  - This is known as the **average decay rate**
- As a result, each radioactive element can be assigned a **decay constant**
- The decay constant λ is defined as:

  **The probability, per second, that a given nucleus will decay**
- When a sample is highly radioactive, this means the number of decays per unit time is very high
  - This suggests it has a high level of **activity**
- Activity, or the number of decays per unit time can be calculated using:

$$A = \frac{\Delta N}{\Delta t} = -\lambda N$$

- Where:
  - $A$ = activity of the sample (Bq)
  - $\Delta N$ = number of decayed nuclei
  - $\Delta t$ = time interval (s)
  - $\lambda$ = decay constant ($s^{-1}$)
  - $N$ = number of nuclei remaining in a sample
- The activity of a sample is measured in **Becquerels** (Bq)
  - An activity of 1 Bq is equal to one decay per second, or 1 $s^{-1}$
- This equation shows:
  - The greater the decay constant, the greater the activity of the sample
  - The activity depends on the number of undecayed nuclei remaining in the sample
  - The minus sign indicates that the number of nuclei remaining decreases with time - however, for calculations it can be omitted

**Worked Example**


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Americium-241 is an artificially produced radioactive element that emits α-particles. A sample of americium-241 of mass 5.1 μg is found to have an activity of $5.9 \times 10^5$ Bq.

(a) Determine the number of nuclei in the sample of americium-241.

(b) Determine the decay constant of americium-241.

**Answer:**

**Part (a)**

**Step 1: Write down the known quantities**

- Mass = 5.1 μg = $5.1 \times 10^{-6}$ g
- Molecular mass of americium = 241
- $N_A$ = Avogadro constant

**Step 2: Write down the equation relating number of nuclei, mass and molecular mass**

$$\text{Number of nuclei} = \frac{\text{mass} \times N_A}{\text{molecular mass}}$$

**Step 3: Calculate the number of nuclei**

$$\textit{number of nuclei} = \frac{(5.1 \times 10^{-6}) \times (6.02 \times 10^{23})}{241} = 1.27 \times 10^{16}$$

**Part (b)**

**Step 1: Write the equation for activity**

**Activity, A = λN**

**Step 2: Rearrange for decay constant λ and calculate the answer**

$$\lambda = \frac{A}{N} = \frac{5.9 \times 10^5}{1.27 \times 10^{16}} = 4.65 \times 10^{-11}\ s^{-1}$$

# Exponential Decay

- In radioactive decay, the number of nuclei falls very rapidly, without ever reaching zero
  - Such a model is known as **exponential decay**
- The graph of number of undecayed nuclei and time has a very distinctive shape

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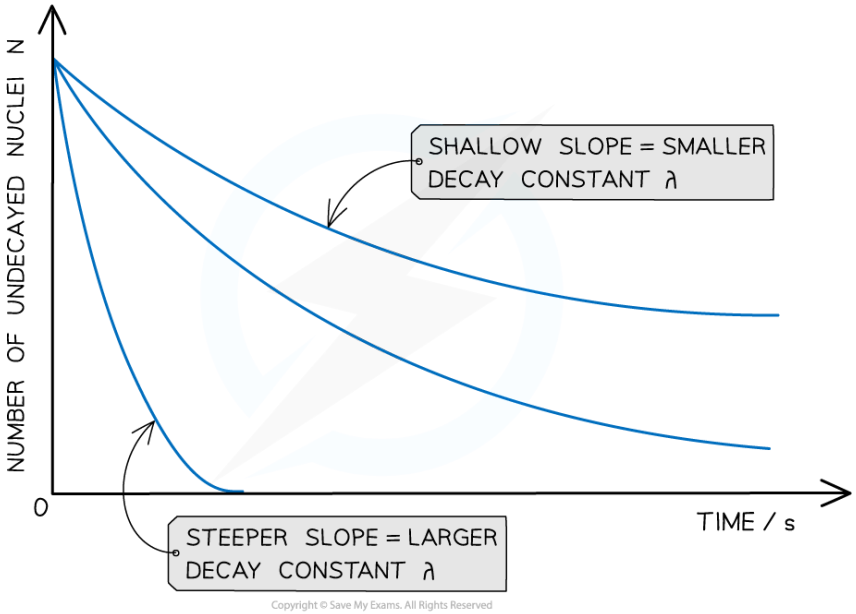


***Radioactive decay follows an exponential pattern. The graph shows three different isotopes each with a different rate of decay***

**Radioactive Decay Equation**

- The number of undecayed nuclei N can be represented in exponential form by the equation:

$$N = N_0 e^{-\lambda t}$$

- Where:
  - $N_0$ = the initial number of undecayed nuclei (when t = 0)
  - $\lambda$ = decay constant ($s^{-1}$)
  - t = time interval (s)

**The exponential function e**

- The symbol e represents the exponential constant
  - It is approximately equal to e = 2.718
- On a calculator it is shown by the button $e^x$
- The inverse function of $e^x$ is ln(y), known as the natural logarithmic function
  - This is because, if $e^x = y$, then x = ln(y)

**Worked Example**

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Strontium-90 decays with the emission of a β-particle to form Yttrium-90. The decay constant of Strontium-90 is 0.025 $year^{-1}$.

Determine the activity A of the sample after 5.0 years, expressing the answer as a fraction of the initial activity $A_0$

**Answer:**

**Step 1: Write out the known quantities**

Decay constant, $\lambda = 0.025\ year^{-1}$

Time interval, t = 5.0 years

Both quantities have the same unit, so there is no need for conversion

**Step 2: Write the equation for activity in exponential form**

$$A = A_0 e^{-\lambda t}$$

**Step 3: Rearrange the equation for the ratio between A and $A_0$**

$$\frac{A}{A_0} = e^{-\lambda t}$$

**Step 4: Calculate the ratio $A/A_0$**

$$\frac{A}{A_0} = e^{-(0.025 \times 5)} = 0.88$$

Therefore, the activity of Strontium-90 decreases by a factor of 0.88, or 12%, after 5 years

# Half Life

- Half-life is defined as:

  **The time taken for half the number of nuclei in a sample to decay**

- This means when a time equal to the half-life has passed, the **activity** of the sample will also half
- This is because the activity is proportional to the number of undecayed nuclei, $A \propto N$

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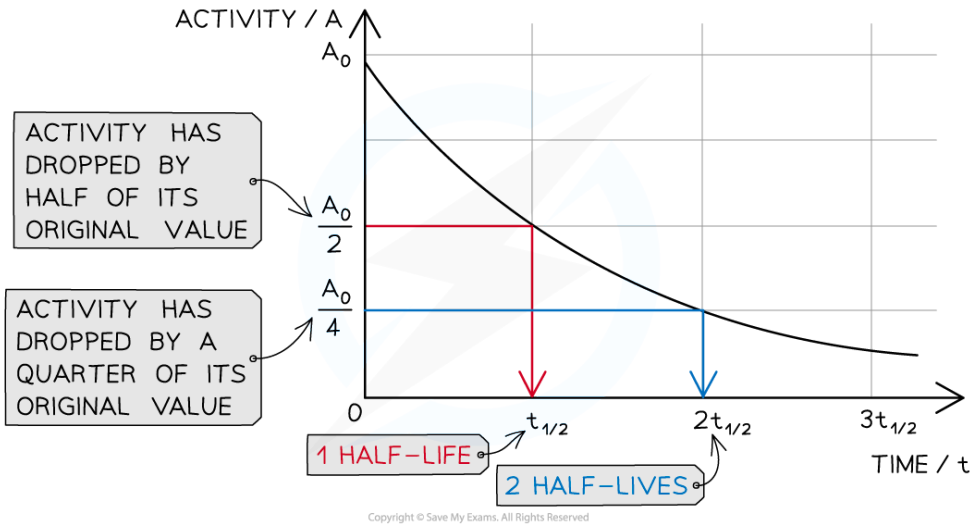


***When a time equal to the half-life passes, the activity falls by half, when two half-lives pass, the activity falls by another half (which is a quarter of the initial value)***

- To find an expression for half-life, start with the equation for exponential decay:

$$N = N_0 e^{-\lambda t}$$

- Where:
  - $N$ = number of nuclei remaining in a sample
  - $N_0$ = the initial number of undecayed nuclei (when t = 0)
  - $\lambda$ = decay constant ($s^{-1}$)
  - $t$ = time interval (s)
- When time $t$ is equal to the half-life $t_{½}$, the activity $N$ of the sample will be half of its original value, so $N = ½ N_0$

$$\frac{1}{2} N_0 = N_0 e^{-\lambda t_{½}}$$

- The formula can then be derived as follows:

Divide both sides by $N_0$: $\frac{1}{2} = e^{-\lambda t_{½}}$

Take the natural log of both sides: $\ln\left(\frac{1}{2}\right) = -\lambda t_{½}$

Apply properties of logarithms: $\lambda t_{½} = \ln(2)$

Your notes

- Therefore, half-life $t_{½}$ can be calculated using the equation:

$$t_{½} = \frac{ln\ 2}{\lambda} \simeq \frac{0.693}{\lambda}$$

- This equation shows that half-life $t_{½}$ and the radioactive decay rate constant λ are inversely proportional
  - Therefore, the **shorter** the half-life, the **larger** the decay constant and the **faster** the decay

Your notes

## Worked Example

Strontium-90 is a radioactive isotope with a half-life of 28.0 years. A sample of Strontium-90 has an activity of $6.4 \times 10^9$ Bq. Calculate the decay constant λ, in $s^{-1}$, of Strontium-90.

**Answer:**

**Step 1: Convert the half-life into seconds**

- $t_{½}$ = 28 years = 28 × 365 × 24 × 60 × 60 = $8.83 \times 10^8$ s

**Step 2: Write the equation for half-life**

$$t_{½} = \frac{ln\ 2}{\lambda}$$

**Step 3: Rearrange for λ and calculate**

$$\lambda = \frac{ln\ 2}{t_{½}} = \frac{ln\ 2}{8.83 \times 10^8} = \mathbf{7.85 \times 10^{-10}\ s^{-1}}$$

## Examiner Tips and Tricks

Although you may not be expected to derive the half-life equation, make sure you're comfortable with how to use it in calculations such as that in the worked example.